

COMP1017 (Mathematics for Computer Scientists) Test 1

Academic Session 2017-2018, Autumn Semester

There are TWO (2) questions pertaining to propositional logic and predicate logic on TWO (2) pages. Please answer ALL the questions. Time allowed 30 minutes.

1. Let p be the proposition "He will come on the 8:15 train", q be "He will come on the 9:15 train", and r be "He will have time to visit us". Given the following propositional formulas that

(1) $(p \vee q) \leftrightarrow r$

(2) $(p \rightarrow r) \wedge (p \vee r)$

(3) $(p \vee \neg q) \rightarrow r$

(4) $(q \leftrightarrow r) \vee (q \leftrightarrow \neg r)$

(a)

p	q	r	$p \vee q$	$(p \vee q) \leftrightarrow r$
0	0	0	0	1
0	0	1	0	0
0	1	0	1	0
0	1	1	1	1
1	0	0	1	0
1	0	1	1	1
1	1	0	1	0
1	1	1	1	1

(a) Find the truth table for (1). $p \ q \ r$

[8 marks]

(b) Show with the logical equivalence laws that (2) can be simplified to r .

$$(p \rightarrow r) \wedge (p \vee r) \Leftrightarrow (\neg p \vee r) \wedge (p \vee r) \Leftrightarrow (\neg p \wedge p) \vee r$$

$$\Leftrightarrow \text{F} \vee r \Leftrightarrow r$$

[10 marks]

(c) Find the contrapositive for (3) and express it in simple English.

$\neg r \rightarrow \neg(p \vee \neg q)$ If he won't have time to visit us, then he won't come on the 8:15 train but will come on the 9:15 train. [10 marks]

(d) Show that (4) is a tautology.

Use either equivalence laws, truth table, or long explanation.

$$(q \leftrightarrow r) \vee (q \leftrightarrow \neg r) \Leftrightarrow (q \rightarrow r) \wedge (r \rightarrow q) \vee (q \rightarrow \neg r) \wedge (\neg r \rightarrow q)$$

$$\Leftrightarrow ((q \vee r) \wedge (\neg q \vee \neg r)) \vee ((q \vee \neg r) \wedge (\neg q \vee r)) \Leftrightarrow (q \vee r) \vee (\neg q \vee \neg r) = \text{T}$$

[12 marks]

(e) Express the sentence "He will come on the 8:15 or the 9:15 train, and if the former, he will have time to visit us" as the propositional formula with p , q , r , and logical operators.

$$(p \vee q) \wedge (p \rightarrow r)$$

[10 marks]

2. Let $G(x, y)$ be the statement " x gets y " and $W(x, y)$ be the statement " x wants y ", respectively. Let the universe of discourse of x be all people, and let the universe of discourse of y be all things that are other than people.

(a) Translate the following sentences into predicate logic notation using $G(x, y)$, $W(x, y)$, and quantifiers.

- (i) If everyone wants something, then no one gets it.

$$\forall y (\forall x W(x, y) \rightarrow \forall x \neg G(x, y)) \quad [15 \text{ marks}]$$

or. $\forall y (\forall x W(x, y) \rightarrow \neg \exists x G(x, y))$

- (ii) There is something that nobody wants.

$$\exists y \forall x \neg \overset{W}{G}(x, y) \quad \text{or.} \quad \exists y \neg \exists x \overset{W}{G}(x, y) \quad [10 \text{ marks}]$$

(b) Translate the predicate logic notation $\exists x \forall y W(x, y) \rightarrow \neg G(x, y)$ to plain English.

Someone ^{only} wants things that they don't get. [15 marks]

(c) If we let the universe of discourse for both x and y be all things and all people, we will need to introduce an additional predicate, in order to translate the sentences in (a) into unambiguous predicate logic notation. What should it be? Explain your work.

It should be $P(x)$: x is a person (or. people) [10 marks]

For example, you will then use $\forall x, y (P(x) \wedge W(x, y) \rightarrow \forall x \neg (P(x) \wedge G(x, y)))$ instead of $\forall x W(x, y) \rightarrow \forall x \neg G(x, y)$.